

MongoDB Replica Set and Sharding Commands

REPLICA SET

```
net stop MongoDB
```

```
mkdir C:\mongo-replica\primary  
mkdir C:\mongo-replica\secondary1  
mkdir C:\mongo-replica\secondary2
```

```
mongod --replSet rs0 --dbpath C:\mongo-replica\primary --port 27017  
mongod --replSet rs0 --dbpath C:\mongo-replica\secondary1 --port 27018  
mongod --replSet rs0 --dbpath C:\mongo-replica\secondary2 --port 27019
```

```
mongosh --port 27017
```

```
rs.initiate({  
  _id: "rs0",  
  members: [  
    { _id: 0, host: "localhost:27017" },  
    { _id: 1, host: "localhost:27018" },  
    { _id: 2, host: "localhost:27019" }  
  ]  
})
```

```
rs.status()
```

```
show dbs
```

SHARDING

```
mkdir C:\mongo-shard  
mkdir C:\mongo-shard\config  
mkdir C:\mongo-shard\shard1  
mkdir C:\mongo-shard\shard2
```

```
mongod --configsvr --replSet configReplSet --dbpath C:\mongo-shard\config --port 27019
```

```
mongosh --port 27019
```

```
rs.initiate({
```

```
_id: "configReplSet",
configsvr: true,
members: [
  { _id: 0, host: "localhost:27019" }
]
})
```

mongod --shardsvr --replSet shard1ReplSet --dbpath C:\mongo-shard\shard1 --port 27011

mongosh --port 27011

```
rs.initiate({
  _id: "shard1ReplSet",
  members: [
    { _id: 0, host: "localhost:27011" }
  ]
})
```

mongod --shardsvr --replSet shard2ReplSet --dbpath C:\mongo-shard\shard2 --port 27012

mongosh --port 27012

```
rs.initiate({
  _id: "shard2ReplSet",
  members: [
    { _id: 0, host: "localhost:27012" }
  ]
})
```

mongos --configdb configReplSet/localhost:27019 --port 27017

mongosh --port 27017

sh.addShard("shard1ReplSet/localhost:27011")

sh.addShard("shard2ReplSet/localhost:27012")

sh.enableSharding("eCommerceDB")

sh.shardCollection("eCommerceDB.orders", { customer_id: "hashed" })

use eCommerceDB

```

for(let i=0;i<1000;i++){
  db.orders.insertOne({
    customer_id:"cust_"+i,
    amount:Math.random()*100
  })
}

```

```
sh.status()
```

```
db.orders.find().limit(5)
```

```
db.orders.countDocuments()
```

```
db.orders.getShardDistribution()
```

```

Command Prompt - mongod x Command Prompt - mongod x Command Prompt - mongod x Command Prompt - mongos x + -
C:\Users\HPVICTUS>mongod --configsvr --replSet configReplSet --dbpath C:\mongo-shard\config --port 27019
'mongod' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\HPVICTUS>cd C:\Program Files\MongoDB\Server\8.3\bin

C:\Program Files\MongoDB\Server\8.3\bin>mongod --configsvr --replSet configReplSet --dbpath C:\mongo-shard\config --port
27019
{"t":{"$date":"2026-06-10T19:25:12.735+05:30"},"s":"I", "c":"-", "id":8991200, "ctx":"thread1", "msg":"Shuffling
initializers", "attr":{"seed":1113957076}}
{"t":{"$date":"2026-06-10T19:25:12.746+05:30"},"s":"I", "c":"CONTROL", "id":97374, "ctx":"thread1", "msg":"Automatica
lly disabling TLS 1.0 and TLS 1.1, to force-enable TLS 1.1 specify --sslDisabledProtocols 'TLS1_0'; to force-enable TLS
1.0 specify --sslDisabledProtocols 'none'}
{"t":{"$date":"2026-06-10T19:25:12.784+05:30"},"s":"I", "c":"EXECUTOR", "id":11280000, "svc":"-", "ctx":"thread1", "msg":
"Starting thread pool", "attr":{"poolName":"AuthorizationRouter", "numThreads":0, "minThreads":0, "maxThreads":1000000000}}
{"t":{"$date":"2026-06-10T19:25:12.785+05:30"},"s":"I", "c":"NETWORK", "id":4915701, "svc":"-", "ctx":"thread1", "msg":
"Initialized wire specification", "attr":{"spec":{"incomingExternalClient":{"minWireVersion":0, "maxWireVersion":28}, "inco
mingInternalClient":{"minWireVersion":0, "maxWireVersion":28}, "outgoing":{"minWireVersion":6, "maxWireVersion":28}, "isInte
rnalClient":true}}}
{"t":{"$date":"2026-06-10T19:25:12.786+05:30"},"s":"I", "c":"EXECUTOR", "id":11280000, "svc":"-", "ctx":"thread1", "msg":
"Starting thread pool", "attr":{"poolName":"ThreadPool1", "numThreads":0, "minThreads":0, "maxThreads":1}}
{"t":{"$date":"2026-06-10T19:25:12.787+05:30"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1", "msg":
"Successfully registered PrimaryOnlyService", "attr":{"service":"ReshardingCoordinatorService", "namespace":"config.reshar
dingOperations"}}
{"t":{"$date":"2026-06-10T19:25:12.787+05:30"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1", "msg":
"Successfully registered PrimaryOnlyService", "attr":{"service":"ConfigsvrCoordinatorService", "namespace":"config.shardin
g_configsvr_coordinators"}}
{"t":{"$date":"2026-06-10T19:25:12.787+05:30"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1", "msg":
"Successfully registered PrimaryOnlyService", "attr":{"service":"ShardingDDLCoordinator", "namespace":"config.system.shard

```

```

Command Prompt - mongod x Command Prompt - mongod x Command Prompt - mongod x Command Prompt - mongos x + -
Microsoft Windows [Version 10.0.26200.8524]
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C:\Users\HPVICTUS>cd C:\Program Files\MongoDB\Server\8.3\bin

C:\Program Files\MongoDB\Server\8.3\bin>mongosh --port 27019
'mongosh' is not recognized as an internal or external command,
operable program or batch file.

C:\Program Files\MongoDB\Server\8.3\bin>mongod --shardsvr --replSet shard1ReplSet --dbpath C:\mongo-shard\shard1 --port
27011
{"t":{"$date":"2026-06-10T19:31:12.828+05:30"},"s":"I", "c":"-", "id":8991200, "ctx":"thread1", "msg":"Shuffling
initializers", "attr":{"seed":2851708836}}
{"t":{"$date":"2026-06-10T19:31:12.843+05:30"},"s":"I", "c":"CONTROL", "id":97374, "ctx":"thread1", "msg":
"Automatically disabling TLS 1.0 and TLS 1.1, to force-enable TLS 1.1 specify --sslDisabledProtocols 'TLS1_0'; to force-
enable TLS 1.0 specify --sslDisabledProtocols 'none'}
{"t":{"$date":"2026-06-10T19:31:12.877+05:30"},"s":"I", "c":"EXECUTOR", "id":11280000, "svc":"-", "ctx":"thread1", "msg":
"Starting thread pool", "attr":{"poolName":"AuthorizationRouter", "numThreads":0, "minThreads":0, "maxThreads":1000000000}}
{"t":{"$date":"2026-06-10T19:31:12.879+05:30"},"s":"I", "c":"NETWORK", "id":4915701, "svc":"-", "ctx":"thread1", "msg":
"Initialized wire specification", "attr":{"spec":{"incomingExternalClient":{"minWireVersion":0, "maxWireVersion":28}, "inco
mingInternalClient":{"minWireVersion":0, "maxWireVersion":28}, "outgoing":{"minWireVersion":6, "maxWireVersion":28}, "isInte
rnalClient":true}}}
{"t":{"$date":"2026-06-10T19:31:12.879+05:30"},"s":"I", "c":"EXECUTOR", "id":11280000, "svc":"-", "ctx":"thread1", "msg":
"Starting thread pool", "attr":{"poolName":"ThreadPool1", "numThreads":0, "minThreads":0, "maxThreads":1}}
{"t":{"$date":"2026-06-10T19:31:12.879+05:30"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1", "msg":
"Successfully registered PrimaryOnlyService", "attr":{"service":"ShardingDDLCoordinator", "namespace":"config.system.shard
ing_ddl_coordinators"}}
{"t":{"$date":"2026-06-10T19:31:12.879+05:30"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1", "msg":
"Successfully registered PrimaryOnlyService", "attr":{"service":"RenameCollectionParticipantService", "namespace":"config.
LocalRenameParticipants"}}

```

```

Microsoft Windows [Version 10.0.26200.8524]
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C:\Users\HPVICTUS>cd C:\Program Files\MongoDB\Server\8.3\bin

C:\Program Files\MongoDB\Server\8.3\bin>mongod --shardsvr --replSet shard2ReplSet --dbpath C:\mongo-shard\shard2 --port
27012
{"t":{"$date":"2026-06-10T19:32:01.906+05:30"},"s":"I", "c":"-", "id":8991200, "ctx":"thread1", "msg":"Shuffling
initializers", "attr":{"seed":537170440}}
{"t":{"$date":"2026-06-10T19:32:01.951+05:30"},"s":"I", "c":"CONTROL", "id":97374, "ctx":"thread1", "msg":"Automatica
lly disabling TLS 1.0 and TLS 1.1, to force-enable TLS 1.1 specify --sslDisabledProtocols 'TLS1_0'; to force-enable TLS
1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2026-06-10T19:32:01.954+05:30"},"s":"I", "c":"EXECUTOR", "id":11280000, "svc":"-", "ctx":"thread1", "msg":
"Starting thread pool", "attr":{"poolName":"AuthorizationRouter", "numThreads":0, "minThreads":0, "maxThreads":1000000000}}
{"t":{"$date":"2026-06-10T19:32:01.954+05:30"},"s":"I", "c":"NETWORK", "id":4915701, "svc":"-", "ctx":"thread1", "msg":
"Initialized wire specification", "attr":{"spec":{"incomingExternalClient":{"minWireVersion":0, "maxWireVersion":28}, "inco
mingInternalClient":{"minWireVersion":0, "maxWireVersion":28}, "outgoing":{"minWireVersion":6, "maxWireVersion":28}, "isInte
rnalClient":true}}}
{"t":{"$date":"2026-06-10T19:32:01.956+05:30"},"s":"I", "c":"EXECUTOR", "id":11280000, "svc":"-", "ctx":"thread1", "msg":
"Starting thread pool", "attr":{"poolName":"ThreadPool1", "numThreads":0, "minThreads":0, "maxThreads":1}}
{"t":{"$date":"2026-06-10T19:32:01.956+05:30"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1", "msg":
"Successfully registered PrimaryOnlyService", "attr":{"service":"ShardingDDLCoordinator", "namespace":"config.system.shard
ing_ddl_coordinators"}}
{"t":{"$date":"2026-06-10T19:32:01.956+05:30"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1", "msg":
"Successfully registered PrimaryOnlyService", "attr":{"service":"RenameCollectionParticipantService", "namespace":"config.
localRenameParticipants"}}
{"t":{"$date":"2026-06-10T19:32:01.956+05:30"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1", "msg":
"Successfully registered PrimaryOnlyService", "attr":{"service":"ReshardingDonorService", "namespace":"config.localReshard
ingOperations.donor"}}
{"t":{"$date":"2026-06-10T19:32:01.956+05:30"},"s":"I", "c":"REPL", "id":5123008, "svc":"-", "ctx":"thread1", "msg":

```

```

Microsoft Windows [Version 10.0.26200.8524]
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C:\Users\HPVICTUS>cd C:\Program Files\MongoDB\Server\8.3\bin

C:\Program Files\MongoDB\Server\8.3\bin>

C:\Program Files\MongoDB\Server\8.3\bin>mongos --configdb configReplSet/localhost:27019 --port 27017
{"t":{"$date":"2026-06-10T19:35:25.755+05:30"},"s":"I", "c":"-", "id":8991200, "ctx":"thread1", "msg":"Shuffling
initializers", "attr":{"seed":2158009861}}
{"t":{"$date":"2026-06-10T19:35:25.778+05:30"},"s":"W", "c":"SHARDING", "id":24132, "ctx":"thread1", "msg":"Running a
sharded cluster with fewer than 3 config servers should only be done for testing purposes and is not recommended for pro
duction."}
{"t":{"$date":"2026-06-10T19:35:25.843+05:30"},"s":"I", "c":"CONTROL", "id":97374, "svc":"-", "ctx":"thread1", "msg":
"Automatically disabling TLS 1.0 and TLS 1.1, to force-enable TLS 1.1 specify --sslDisabledProtocols 'TLS1_0'; to force-
enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2026-06-10T19:35:25.848+05:30"},"s":"I", "c":"EXECUTOR", "id":11280000, "svc":"-", "ctx":"thread1", "msg":
"Starting thread pool", "attr":{"poolName":"AuthorizationRouter", "numThreads":0, "minThreads":0, "maxThreads":1000000000}}
{"t":{"$date":"2026-06-10T19:35:25.855+05:30"},"s":"I", "c":"NETWORK", "id":4915701, "svc":"-", "ctx":"thread1", "msg":
"Initialized wire specification", "attr":{"spec":{"incomingExternalClient":{"minWireVersion":0, "maxWireVersion":28}, "inco
mingInternalClient":{"minWireVersion":0, "maxWireVersion":28}, "outgoing":{"minWireVersion":28, "maxWireVersion":28}, "isInt
ernalClient":true}}}
{"t":{"$date":"2026-06-10T19:35:25.857+05:30"},"s":"I", "c":"HEALTH", "id":5936503, "svc":"-", "ctx":"thread1", "msg":
"Fault manager changed state", "attr":{"state":"StartupCheck"}}
{"t":{"$date":"2026-06-10T19:35:25.858+05:30"},"s":"W", "c":"CONTROL", "id":22120, "svc":"-", "ctx":"thread1", "msg":
"Access control is not enabled for the database. Read and write access to data and configuration is unrestricted", "tags"
:["startupWarnings"]}
{"t":{"$date":"2026-06-10T19:35:25.858+05:30"},"s":"W", "c":"CONTROL", "id":22140, "svc":"-", "ctx":"thread1", "msg":
"This server is bound to localhost. Remote systems will be unable to connect to this server. Start the server with --bin
d_ip <address> to specify which IP addresses it should serve responses from, or with --bind_ip_all to bind to all interf

```

The screenshot shows the MongoDB Compass web interface. The top navigation bar includes 'Welcome', 'mongodb: learning', 'mongodb: learning2', 'mongodb: localhost:27012', and 'mongodb: localhost:27017'. The left sidebar shows a tree view of connections, including 'learning', 'learning2', 'localhost:27012', and 'localhost:27017'. The main panel displays the 'shardingVersion' configuration for the 'localhost:27017' connection. The configuration shows a sharded cluster with two shards: 'shard1ReplSet' and 'shard2ReplSet'. Each shard is a replica set with a primary member. The 'shard1ReplSet' has a primary member at 'localhost:27011' and a secondary member at 'localhost:27012'. The 'shard2ReplSet' has a primary member at 'localhost:27012' and a secondary member at 'localhost:27017'. The 'shard1ReplSet' is configured with a topology time of '1781100358' and a replication set configuration version of '-1'. The 'shard2ReplSet' is configured with a topology time of '1781100377' and a replication set configuration version of '-1'. The 'shard1ReplSet' is also configured with a primary member at 'localhost:27011' and a secondary member at 'localhost:27012'. The 'shard2ReplSet' is also configured with a primary member at 'localhost:27012' and a secondary member at 'localhost:27017'.

Output MongoDBsh :

```
sh.addShard("shard1ReplSet/localhost:27011")
```

```
{
  shardAdded: 'shard1ReplSet',
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1781100359, i: 3 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1781100359, i: 3 })
}
```

```
sh.addShard("shard2ReplSet/localhost:27012")
```

```
{
  shardAdded: 'shard2ReplSet',
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1781100377, i: 57 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1781100377, i: 57 })
}
```

```

}

sh.enableSharding("eCommerceDB")

{
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1781100390, i: 7 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1781100390, i: 7 })
}

sh.shardCollection(
  "eCommerceDB.orders",
  { customer_id: "hashed" }
)

{
  collectionssharded: 'eCommerceDB.orders',
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1781100403, i: 48 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  }
}

```

```
},  
  operationTime: Timestamp({ t: 1781100403, i: 47 })  
}
```

use eCommerceDB

```
for(let i=0;i<1000;i++){  
  db.orders.insertOne({  
    customer_id:"cust_"+i,  
    amount:Math.random()*100  
  })  
}
```

switched to db eCommerceDB

```
for(let i=0;i<1000;i++){  
  db.orders.insertOne({  
    customer_id:"cust_"+i,  
    amount:Math.random()*100  
  })  
}  
  
{  
  acknowledged: true,  
  insertedId: ObjectId('6a296fc5f27f8e15b701c459')  
}
```

sh.status()

shardingVersion

```
{  
  _id: ObjectId('6a296e1c562bc466b5b94c5c'),
```

```
clusterId: ObjectId('6a296e1c562bc466b5b94c5b')
}
shards
[
  {
    _id: 'shard1ReplSet',
    host: 'shard1ReplSet/localhost:27011',
    topologyTime: Timestamp({ t: 1781100358, i: 32 }),
    replSetConfigVersion: -1
  },
  {
    _id: 'shard2ReplSet',
    host: 'shard2ReplSet/localhost:27012',
    topologyTime: Timestamp({ t: 1781100377, i: 33 }),
    replSetConfigVersion: -1
  }
]
active mongoses
[
  {
    '8.3.2': 1
  }
]
autosplit
{
  'Currently enabled': 'yes'
```



```
}  
  
balancer  
  
{  
  'Currently enabled': 'yes',  
  'Currently running': 'no',  
  'Failed balancer rounds in last 5 attempts': 0,  
  'Migration Results for the last 24 hours': 'No recent migrations'  
}  
  
shardedDataDistribution  
  
[  
  {  
    ns: 'eCommerceDB.orders',  
    shards: [  
      {  
        shardName: 'shard2ReplSet',  
        numOrphanedDocs: 0,  
        numOwnedDocuments: 509,  
        ownedSizeBytes: 32067,  
        orphanedSizeBytes: 0  
      },  
      {  
        shardName: 'shard1ReplSet',  
        numOrphanedDocs: 0,  
        numOwnedDocuments: 491,  
        ownedSizeBytes: 30933,  
        orphanedSizeBytes: 0  
      }  
    ]  
  }  
]
```

```
    }
  ]
}
]
databases
[
  collections: {}
},
{
  database: {
    _id: 'eCommerceDB',
    primary: 'shard1ReplSet',
    version: {
      uuid: UUID('cad5c703-d634-4958-b374-c60e1625cbd9'),
      timestamp: Timestamp({ t: 1781100389, i: 10 }),
      lastMod: 1
    }
  },
  collections: {
    'eCommerceDB.orders': {
      shardKey: {
        customer_id: 'hashed'
      },
      unique: false,
      balancing: true,
      allowMigrations: true,
```

```
chunkMetadata: [  
  {  
    shard: 'shard1ReplSet',  
    nChunks: 1  
  },  
  {  
    shard: 'shard2ReplSet',  
    nChunks: 1  
  }  
],  
chunks: [  
  {  
    min: {  
      customer_id: MinKey()  
    },  
    max: {  
      customer_id: 0  
    },  
    'on shard': 'shard2ReplSet',  
    'last modified': Timestamp({ t: 1, i: 0 })  
  },  
  {  
    min: {  
      customer_id: 0  
    },  
    max: {
```

```

        customer_id: MaxKey()
    },
    'on shard': 'shard1ReplSet',
    'last modified': Timestamp({ t: 1, i: 1 })
}
],
tags: []
}
}
}
]
db.orders.find().limit(5)
{
  _id: ObjectId('6a296fbaf27f8e15b701c072'),
  customer_id: 'cust_0',
  amount: 22.496351071277054
}
{
  _id: ObjectId('6a296fbaf27f8e15b701c077'),
  customer_id: 'cust_5',
  amount: 77.08264499001638
}
{
  _id: ObjectId('6a296fbaf27f8e15b701c078'),
  customer_id: 'cust_6',
  amount: 79.86581795700384
}

```

```
}
```

```
{
```

```
  _id: ObjectId('6a296fbaf27f8e15b701c079'),
```

```
  customer_id: 'cust_7',
```

```
  amount: 15.785361886288696
```

```
}
```